



11054 - Breaking Degeneracies in Protoplanetary Disk CO₂ Retrievals with NIRSpec

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	HOTau Obs	NIRSpec IFU Spectroscopy	(1) V-HO-Tau
	2	V836Tau Obs	NIRSpec IFU Spectroscopy	(3) V-V836-Tau
	3	Sz-66 Obs	NIRSpec IFU Spectroscopy	(4) Sz-66
	4	Sz-77 Obs	NIRSpec IFU Spectroscopy	(5) Sz-77
	5	Sz-90 Obs	NIRSpec IFU Spectroscopy	(6) Sz90
	6	Sz-95 Obs	NIRSpec IFU Spectroscopy	(7) Sz-95
	7	Sz-96 Obs	NIRSpec IFU Spectroscopy	(8) Sz-96

Folder	Observation	Label	Observing Template	Science Target
	8	Sz-118 Obs	NIRSpec IFU Spectroscopy	(9) Sz-118
	9	Sz-123a Obs	NIRSpec IFU Spectroscopy	(10) Sz123a
	10	Sz-130 Obs	NIRSpec IFU Spectroscopy	(11) Sz-130
	11	J1612 Obs	NIRSpec IFU Spectroscopy	(12) J1612
	12	FNTau Obs	NIRSpec IFU Spectroscopy	(13) V-FN-Tau
	13	Sz-65 Obs	NIRSpec IFU Spectroscopy	(14) Sz-65
	14	DSTau Obs	NIRSpec IFU Spectroscopy	(15) V-DS-Tau

ABSTRACT

We propose JWST NIRSpec IFU observations of the 4.3 micron CO₂ v₃ band and the 4.5-5 micron CO rovibrational band in 14 protoplanetary disks that already have JWST MIRI spectra. Disks set the initial conditions for planet formation, so robust constraints on molecular inventories in the inner, planet-forming region within 10 au are essential. Analyses based on MIRI alone suffer from strong degeneracies among temperature, column density, and emitting area, driven by limited transition coverage and line blending. NIRSpec provides key complementary transitions that respond sensitively to the same underlying parameters, allowing us to distinguish optically thick from optically thin CO₂ emission. By resolving the R and P branches of CO₂ and sampling the CO rovibrational comb, we will jointly fit NIRSpec and MIRI spectra to reduce uncertainties on temperature and column density by factors of order two across a representative set of compact and extended disks. The resulting constraints on the distribution and abundances of major C- and O-bearing species will strengthen links between disk chemistry and exoplanet compositions.

OBSERVING DESCRIPTION

We propose to use JWST NIRSpec IFU to observe 14 disks targeted as part of the ALMA Disk-Exoplanet C/Onnection Large Program, which has already acquired millimeter (ALMA) and mid-infrared (MIRI) data on these objects. We aim to observe infrared CO₂ and CO lines to better trace the inner, planet-forming regions of disks and break degeneracies present in MIRI observations. Our lines of interest span from 3 - 5.15 microns, and so we will use the F290LP filter. Though full line profiles are unresolved, we require R~2700 to resolve individual line peaks, and therefore we will use the G395H grating. Based on our ETC calculations and previous conversations with the Help Desk, we adopt a NRSRAPID readout pattern and will dither 10 times in a cycling pattern. The brightness of the targets at 4.3 microns sets the exact ratio of groups and integrations to optimize for cosmic ray mitigation or avoiding saturation.

Proposal 11054 - Targets - Breaking Degeneracies in Protoplanetary Disk CO2 Retrievals with NIRSpect

(7)	Sz-95	RA: 16 07 52.3110 (241.9679625d) Dec: -38 58 6.09 (-38.96836d) Equinox: J2000	Proper Motion RA: -10.607 mas/yr Proper Motion Dec: -23.424999949384073 mas/yr Parallax: 0.0062317" Epoch of Position: 2000
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Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.

Category=Star

Description=[Circumstellar gas, Protoplanetary disks]

(8)	Sz-96	RA: 16 08 12.6309 (242.0526287d) Dec: -39 08 33.47 (-39.14263d) Equinox: J2000	Proper Motion RA: -10.251 mas/yr Proper Motion Dec: -23.36899999590969 mas/yr Parallax: 0.0064113" Epoch of Position: 2000
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Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.

Category=Star

Description=[Circumstellar gas, Protoplanetary disks]

(9)	Sz-118	RA: 16 09 48.6562 (242.4527342d) Dec: -39 11 16.85 (-39.18801d) Equinox: J2000	Proper Motion RA: -10.573 mas/yr Proper Motion Dec: -23.569999916617235 mas/yr Parallax: 0.0061935" Epoch of Position: 2000
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Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.

Category=Star

Description=[Circumstellar gas, Protoplanetary disks]

(10)	Sz123a	RA: 16 10 51.5850 (242.7149375d) Dec: -38 53 13.77 (-38.88716d) Equinox: J2000	Proper Motion RA: -8.992 mas/yr Proper Motion Dec: -23.38100007364119 mas/yr Parallax: 0.0061656" Epoch of Position: 2000
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Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.

Category=Star

Description=[Circumstellar gas, Protoplanetary disks]

(11)	Sz-130	RA: 16 00 31.0375 (240.1293229d) Dec: -41 43 36.99 (-41.72694d) Equinox: J2000	Proper Motion RA: -10.845 mas/yr Proper Motion Dec: -23.3380000508987 mas/yr Parallax: 0.0062822" Epoch of Position: 2000
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Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.

Category=Star

Description=[Circumstellar gas, Protoplanetary disks]

Proposal 11054 - Targets - Breaking Degeneracies in Protoplanetary Disk CO2 Retrievals with NIRSpec

(12)	J1612	RA: 16 12 43.7499 (243.1822912d) Dec: -38 15 3.08 (-38.25086d) Equinox: J2000	Proper Motion RA: -8.499 mas/yr Proper Motion Dec: -22.412999942389433 mas/yr Parallax: 0.0062559" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Circumstellar gas, Protoplanetary disks]			
(13)	V-FN-Tau	RA: 04 14 14.5945 (63.5608104d) Dec: +28 27 58.07 (28.46613d) Equinox: J2000	Proper Motion RA: 9.344 mas/yr Proper Motion Dec: -23.83900007316697 mas/yr Parallax: 0.0076986" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Circumstellar gas, Protoplanetary disks]			
(14)	Sz-65	RA: 15 39 27.7717 (234.8657154d) Dec: -34 46 17.21 (-34.77145d) Equinox: J2000	Proper Motion RA: -13.218 mas/yr Proper Motion Dec: -22.121000097286014 mas/yr Parallax: 0.0065159" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Circumstellar gas, Protoplanetary disks]			
(15)	V-DS-Tau	RA: 04 47 48.5955 (71.9524813d) Dec: +29 25 11.19 (29.41978d) Equinox: J2000	Proper Motion RA: 5.049 mas/yr Proper Motion Dec: -24.439000071652117 mas/yr Parallax: 0.006315" Epoch of Position: 2000
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.</i> Category=Star Description=[Circumstellar gas, Protoplanetary disks]			

Proposal 11054 - Observation 1 - Breaking Degeneracies in Protoplanetary Disk CO2 Retrievals with NIRSpec

Observation	Proposal 11054, Observation 1: HOTau Obs											Fri Mar 13 20:05:17 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	V-HO-Tau	RA: 04 35 20.2105 (68.8342104d) Dec: +22 32 14.56 (22.53738d) Equinox: J2000				Proper Motion RA: 9.721 mas/yr Proper Motion Dec: -16.898000080800557 mas/yr Parallax: 0.0060796" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Circumstellar gas, Protoplanetary disks]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		MEDIUM		5		10				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	4	3	false	true	NONE	10	30	2188.333	

Proposal 11054 - Observation 2 - Breaking Degeneracies in Protoplanetary Disk CO2 Retrievals with NIRSpec

Observation	Proposal 11054, Observation 2: V836Tau Obs											Fri Mar 13 20:05:17 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	V-V836-Tau		RA: 05 03 6.5988 (75.7774950d) Dec: +25 23 19.61 (25.38878d) Equinox: J2000			Proper Motion RA: 2.991 mas/yr Proper Motion Dec: -17.303999948126148 mas/yr Parallax: 0.0059882" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Circumstellar gas, Protoplanetary disks]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			MEDIUM		5		10			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSRAPID	2	5	false	true	NONE	10	50	1610.515	

Proposal 11054 - Observation 3 - Breaking Degeneracies in Protoplanetary Disk CO2 Retrievals with NIRSpec

Observation	Proposal 11054, Observation 3: Sz-66 Obs											Fri Mar 13 20:05:17 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(4)	Sz-66	RA: 15 39 28.2829 (234.8678454d) Dec: -34 46 18.08 (-34.77169d) Equinox: J2000				Proper Motion RA: -13.124 mas/yr Proper Motion Dec: -21.80700000735669 mas/yr Parallax: 0.006413400000000005" Epoch of Position: 2000			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Circumstellar gas, Protoplanetary disks]		
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			MEDIUM		5		10			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSRAPID	3	4	false	true	NONE	10	40	1717.883	

Proposal 11054 - Observation 4 - Breaking Degeneracies in Protoplanetary Disk CO2 Retrievals with NIRSpec

Observation	Proposal 11054, Observation 4: Sz-77 Obs											Fri Mar 13 20:05:17 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(5)	Sz-77		RA: 15 51 46.9581 (237.9456587d) Dec: -35 56 44.11 (-35.94559d) Equinox: J2000			Proper Motion RA: -12.55 mas/yr Proper Motion Dec: -24.183000050470582 mas/yr Parallax: 0.0064411" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Circumstellar gas, Protoplanetary disks]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			MEDIUM		5		10			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSRAPID	2	5	false	true	NONE	10	50	1610.515	

Proposal 11054 - Observation 5 - Breaking Degeneracies in Protoplanetary Disk CO2 Retrievals with NIRSpec

Observation	Proposal 11054, Observation 5: Sz-90 Obs											Fri Mar 13 20:05:17 GMT 2026
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous	
	(6)	Sz90	RA: 16 07 10.0694 (241.7919558d) Dec: -39 11 3.26 (-39.18424d) Equinox: J2000				Proper Motion RA: -9.353 mas/yr Proper Motion Dec: -25.295000091318798 mas/yr Parallax: 0.006235699999999999" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Circumstellar gas, Protoplanetary disks]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			MEDIUM		5		10			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSRAPID	2	5	false	true	NONE	10	50	1610.515	

Proposal 11054 - Observation 6 - Breaking Degeneracies in Protoplanetary Disk CO2 Retrievals with NIRSpec

Observation	Proposal 11054, Observation 6: Sz-95 Obs												Fri Mar 13 20:05:17 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(7)	Sz-95	RA: 16 07 52.3110 (241.9679625d) Dec: -38 58 6.09 (-38.96836d) Equinox: J2000				Proper Motion RA: -10.607 mas/yr Proper Motion Dec: -23.424999949384073 mas/yr Parallax: 0.0062317" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Circumstellar gas, Protoplanetary disks]												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	CYCLING			MEDIUM		5		10				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G395H/F290LP	NRSIRS2RAPID	7	2	false	true	NONE	10	20	2334.222		

Proposal 11054 - Observation 7 - Breaking Degeneracies in Protoplanetary Disk CO2 Retrievals with NIRSpec

Observation	Proposal 11054, Observation 7: Sz-96 Obs											Fri Mar 13 20:05:17 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(8)	Sz-96	RA: 16 08 12.6309 (242.0526287d) Dec: -39 08 33.47 (-39.14263d) Equinox: J2000				Proper Motion RA: -10.251 mas/yr Proper Motion Dec: -23.36899999590969 mas/yr Parallax: 0.0064113" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Circumstellar gas, Protoplanetary disks]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			MEDIUM		5		10			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSRAPID	2	5	false	true	NONE	10	50	1610.515	

Proposal 11054 - Observation 8 - Breaking Degeneracies in Protoplanetary Disk CO2 Retrievals with NIRSpec

Observation	Proposal 11054, Observation 8: Sz-118 Obs											Fri Mar 13 20:05:17 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(9)	Sz-118	RA: 16 09 48.6562 (242.4527342d) Dec: -39 11 16.85 (-39.18801d) Equinox: J2000				Proper Motion RA: -10.573 mas/yr Proper Motion Dec: -23.569999916617235 mas/yr Parallax: 0.0061935" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Circumstellar gas, Protoplanetary disks]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			MEDIUM		5		10			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSRAPID	2	5	false	true	NONE	10	50	1610.515	

Proposal 11054 - Observation 9 - Breaking Degeneracies in Protoplanetary Disk CO2 Retrievals with NIRSpec

Observation	Proposal 11054, Observation 9: Sz-123a Obs											Fri Mar 13 20:05:17 GMT 2026
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(10)	Sz123a	RA: 16 10 51.5850 (242.7149375d) Dec: -38 53 13.77 (-38.88716d) Equinox: J2000				Proper Motion RA: -8.992 mas/yr Proper Motion Dec: -23.38100007364119 mas/yr Parallax: 0.0061656" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Circumstellar gas, Protoplanetary disks]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		MEDIUM		5		10				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	7	2	false	true	NONE	10	20	2334.222	

Proposal 11054 - Observation 10 - Breaking Degeneracies in Protoplanetary Disk CO2 Retrievals with NIRSpec

Observation	Proposal 11054, Observation 10: Sz-130 Obs											Fri Mar 13 20:05:17 GMT 2026
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(11)	Sz-130	RA: 16 00 31.0375 (240.1293229d) Dec: -41 43 36.99 (-41.72694d) Equinox: J2000				Proper Motion RA: -10.845 mas/yr Proper Motion Dec: -23.3380000508987 mas/yr Parallax: 0.0062822" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Circumstellar gas, Protoplanetary disks]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			MEDIUM		5		10			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSRAPID	3	4	false	true	NONE	10	40	1717.883	

Proposal 11054 - Observation 11 - Breaking Degeneracies in Protoplanetary Disk CO2 Retrievals with NIRSpec

Observation	Proposal 11054, Observation 11: J1612 Obs											Fri Mar 13 20:05:17 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(12)	J1612	RA: 16 12 43.7499 (243.1822912d) Dec: -38 15 3.08 (-38.25086d) Equinox: J2000				Proper Motion RA: -8.499 mas/yr Proper Motion Dec: -22.412999942389433 mas/yr Parallax: 0.0062559" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Circumstellar gas, Protoplanetary disks]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			MEDIUM		5		10			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSRAPID	4	3	false	true	NONE	10	30	1610.515	

Proposal 11054 - Observation 12 - Breaking Degeneracies in Protoplanetary Disk CO2 Retrievals with NIRSpec

Observation	Proposal 11054, Observation 12: FNTau Obs											Fri Mar 13 20:05:17 GMT 2026
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(FNTau Obs (Obs 12)) Warning (Form): NGROUPS=1 may suffer from low calibration accuracy. (Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(13)	V-FN-Tau	RA: 04 14 14.5945 (63.5608104d) Dec: +28 27 58.07 (28.46613d) Equinox: J2000				Proper Motion RA: 9.344 mas/yr Proper Motion Dec: -23.83900007316697 mas/yr Parallax: 0.0076986" Epoch of Position: 2000					
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
Category=Star												
Description=[Circumstellar gas, Protoplanetary disks]												
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			MEDIUM		5		10			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSRAPID	1	8	false	true	NONE	10	80	1717.883	

Proposal 11054 - Observation 13 - Breaking Degeneracies in Protoplanetary Disk CO2 Retrievals with NIRSpec

Observation	Proposal 11054, Observation 13: Sz-65 Obs											Fri Mar 13 20:05:17 GMT 2026
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Sz-65 Obs (Obs 13)) Warning (Form): NGROUPS=1 may suffer from low calibration accuracy.											
	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(14)	Sz-65	RA: 15 39 27.7717 (234.8657154d) Dec: -34 46 17.21 (-34.77145d) Equinox: J2000				Proper Motion RA: -13.218 mas/yr Proper Motion Dec: -22.121000097286014 mas/yr Parallax: 0.0065159" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Circumstellar gas, Protoplanetary disks]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			MEDIUM		5		10			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSRAPID	1	8	false	true	NONE	10	80	1717.883	

Proposal 11054 - Observation 14 - Breaking Degeneracies in Protoplanetary Disk CO2 Retrievals with NIRSpec

Observation	Proposal 11054, Observation 14: DSTau Obs											Fri Mar 13 20:05:17 GMT 2026	
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(DSTau Obs (Obs 14)) Warning (Form): NGROUPS=1 may suffer from low calibration accuracy.												
	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(15)	V-DS-Tau		RA: 04 47 48.5955 (71.9524813d) Dec: +29 25 11.19 (29.41978d) Equinox: J2000			Proper Motion RA: 5.049 mas/yr Proper Motion Dec: -24.439000071652117 mas/yr Parallax: 0.006315" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Star Description=[Circumstellar gas, Protoplanetary disks]												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points		
	1	CYCLING			MEDIUM		5		10				
Spectral Elements	#	Grating/Filter		Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP		NRSRAPID	1	8	false	true	NONE	10	80	1717.883	